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RESEARCH ARTICLE

When Does Posterior Integration Change Option Values? Evidence from WTI Average Price Options

Anonymous Author[1]

Abstract

We ask whether accounting for volatility parameter uncertainty materially changes WTI Average Price Option (APO) prices, and whether that effect is more important than the volatility information used for valuation. Posterior-integrated pricing (PI) averages option values over the volatility posterior; posterior-mean plug-in pricing (PM) uses its mean. We relate their difference to posterior variance and option-price curvature and examine the mechanism in 10 million synthetic return histories. For 310 October-2026-expiry contract-date observations, the mean absolute PI–PM difference is \$0.000745 per barrel, despite a mean 95% posterior model-price interval width of \$0.2733. In a separate date-ordered conditional pricing comparison involving 2,381 option-date observations on only 15 valuation dates, earlier option-implied volatility produces lower errors against end-of-day settlement-price proxies than the tested historical-volatility estimates (MAE \$0.1075 versus \$2.6187 per barrel). The result also holds on the matched same-contract implied-volatility benchmark, without fitting a smile. Our comparisons are conditional on contemporaneous futures inputs, a one-factor pricing model, sparse trading, and unverified vendor publication times. A small integration correction should not be interpreted as evidence of small parameter-induced price uncertainty or overall pricing accuracy.

KEYWORDS

Bayesian option pricing; Asian options; WTI crude oil; parameter uncertainty; implied volatility

1 | INTRODUCTION

Bayesian option valuation can account for parameter uncertainty by integrating conditional prices over a posterior distribution rather than using a point estimate (Guidolin and Timmermann 2003, Rombouts and Stentoft 2014, Gupta and Reisinger 2014). Whether this integration materially changes the option price depends on the pricing function and the dispersion of the parameter posterior. We study this question for WTI Average Price Options (APOs), focusing on uncertainty about volatility estimated from historical returns.

Arithmetic-average Asian option values are sensitive to moneyness, maturity, and the proportion of the average already fixed. Established methods exploit geometric-average control variates or conditioning approximations (Kemna and Vorst 1990, Curran 1994). Commodity applications must also account for the futures term structure; richer models can incorporate stochastic convenience yields, jumps, and stochastic volatility (Schwartz 1997, Shiraya and Takahashi 2011, Ewald et al. 2023). We instead hold a one-factor pricing model fixed to distinguish two effects: posterior integration of historical-volatility uncertainty and the choice between historical and option-implied volatility inputs.

These effects address different questions. The posterior-integrated price (PI) averages conditional option values over the volatility posterior, whereas the posterior-mean plug-in price (PM) uses posterior mean volatility. Their difference measures the effect of integration on the point price, not the full dispersion of model prices caused by parameter uncertainty. The volatility-input comparison asks a separate empirical question: which estimates give smaller errors

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against settlement-price proxies under the same pricing model? Because repricing uses the valuation-date futures curve and already known fixings, the exercise is not a fully prospective trading or price forecast.

We investigate the PI–PM difference with 10 million synthetic return histories and October-2026 WTI APO observations. We then compare historical-volatility benchmarks with implied volatility estimated from earlier APO dates in a date-ordered conditional pricing panel with 15 valuation dates. As a cross-instrument check, we obtain implied volatility from standard monthly WTI futures options instead of APOs. Implied-volatility estimates use only earlier option observations; repricing uses the valuation-date futures curve, realized fixings, and discount factor.

The synthetic results show that posterior integration can matter when historical samples are short and the pricing function has substantial curvature in volatility. In the observed October-2026 sample, PI and PM are nearly identical despite much wider posterior dispersion in model prices. Earlier-date option-implied volatility also yields lower out-of-sample pricing errors than the tested historical benchmarks, including rolling windows, EWMA, and horizon-matched GARCH(1,1). This advantage persists without fitting a volatility smile and in the cross-instrument check.

This differs from model-complexity comparisons involving alternative stochastic-volatility and jump specifications, such as Cummins and Esposito (2025). We hold the pricing model fixed and examine the consequences of volatility estimation, posterior averaging, and parameter-induced uncertainty for an arithmetic-average commodity option. These are conditional comparisons within a maintained pricing model, not evidence of an executable trading advantage or a structural difference between physical and risk-neutral volatility. The methods and sample restrictions are described below; detailed numerical and robustness results appear in the online supplement.

2 | POSTERIOR UNCERTAINTY AND ASIAN-OPTION PRICING

2.1 | Physical inference and risk-neutral valuation

Historical log returns are modeled under the physical measure as

$$R_i \sim \mathcal{N}\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t, \sigma^2 \Delta t\right], \tag{1}$$

with baseline priors

$$\mu \sim \mathcal{N}(0,1), \qquad \sigma \sim \text{InvGamma}(2, 0.1), \qquad \sigma > 0. \tag{2}$$

We sample the empirical posterior in $(\mu, \log\sigma)$ using random-walk Metropolis. For the large synthetic experiment, we integrate out μ analytically conditional on σ. This leaves a one-dimensional posterior calculation and avoids MCMC error in that experiment.

Pricing is carried out separately under the risk-neutral measure. For futures maturity u, the maintained benchmark is

$$dF_t^u = \sigma F_t^u \, dW_t^Q, \tag{3}$$

where u indexes futures maturity and the same Brownian motion W^Q drives all maturities. The common factor specifies perfect instantaneous correlation across futures maturities under the maintained model. Physical drift does not enter risk-neutral valuation. Holding these dynamics fixed allows direct comparisons of posterior integration and volatility inputs.

For an arithmetic-average option with fixing dates $t_1, \ldots, t_M$,

$$A_T = \frac{1}{M}\sum_{j=1}^{M} F_{t_j}^{(1)}, \tag{4}$$

and the conditional call value is

$$C_t^Q(\sigma) = B(t,T)\,\mathbb{E}^Q\left[(A_T - K)^+ \mid \mathcal{F}_t; \sigma\right], \tag{5}$$

where $B(t,T)$ is the discount factor, K the strike, and $\mathcal{F}_t$ the valuation-date information set. Put values are defined analogously. In Equation (4), $F_{t_j}^{(1)}$ is the first-nearby WTI futures settlement on fixing date t_j. At a valuation date within the averaging period, earlier fixings are known. We simulate only the remaining fixings, using the corresponding contracts in the observed CL futures curve as initial values.

We price the baseline historical-volatility model by Monte Carlo. For the date-ordered conditional comparisons and checks using alternative constant volatilities, we use Curran's deterministic conditioning approximation (Curran 1994). We check numerical accuracy separately with high-precision pseudo-random Monte Carlo and scrambled Sobol calculations.

2.2 | Posterior integration and price uncertainty

Let $\mathcal{D}$ denote the historical return data, $C^Q(\sigma)$ the conditional risk-neutral option value, and $\bar{\sigma} = \mathbb{E}[\sigma \mid \mathcal{D}]$. We

compare posterior-integrated (PI) and posterior-mean plug-in (PM) prices:

$$C_{PI} = \mathbb{E}[C^Q(\sigma) \mid \mathcal{D}], \qquad C_{PM} = C^Q(\bar{\sigma}). \tag{6}$$

A second-order expansion around $\bar{\sigma}$ gives

$$C_{PI} - C_{PM} \approx \frac{1}{2} C^{Q\prime\prime}(\bar{\sigma}) \operatorname{Var}(\sigma \mid \mathcal{D}). \tag{7}$$

The approximation links the PI–PM difference to posterior variance and price curvature. Its sign depends on local curvature, and it is most accurate when higher-order terms are small over the posterior support.

The posterior distribution also induces a distribution of theoretical option values,

$$C^Q(\sigma), \qquad \sigma \sim p(\sigma \mid \mathcal{D}). \tag{8}$$

Its dispersion is approximately

$$\operatorname{SD}\big(C^Q(\sigma) \mid \mathcal{D}\big) \approx \big|C^{Q\prime}(\bar{\sigma})\big| \sqrt{\operatorname{Var}(\sigma \mid \mathcal{D})}. \tag{9}$$

The two quantities have different local orders in posterior dispersion. The PI–PM difference depends on posterior variance, whereas the standard deviation of conditional model prices depends on the posterior standard deviation of volatility. Neither quantity measures future market-price uncertainty or removes pricing-model misspecification.

2.3 | Synthetic mechanism experiment

The synthetic experiment is designed to vary the two ingredients in Equation (7). It contains 10 million simulated return histories, generated with root seed 2468 and distributed over 35 combinations of historical sample size (21 to 1,260 observations) and annual volatility (0.10 to 0.80). We evaluate 168 option states varying maturity, moneyness, and the fraction of the arithmetic average already fixed. For each history and option state, we calculate the PI–PM difference, posterior volatility variance, local price curvature, and the second-order approximation. The volatility posterior is evaluated on a 4,097-point grid; computations are batched without changing the model or pricing grid.

This design isolates the conditions under which posterior integration matters and tests the curvature–variance approximation independently of market pricing errors.

3 | DATA AND EMPIRICAL DESIGN

3.1 | WTI Average Price Options

The empirical sample is built from individual Barchart histories for WTI Average Price Options. The raw archive contains 213 unique contracts and 11,179 contract-date observations, with expiry months ranging from September 2026 to June 2029. We compare model prices with the Barchart end-of-day `Latest` field, used as a settlement-price proxy because official final settlement and trade-execution status have not been independently verified. Calls and puts are retained as available, and moneyness is recomputed by valuation date rather than assigned once at the contract level.

For every observation, we identify the fixings already known and the futures contracts used for the remaining fixings. We initialize those contracts with the futures prices observed on the valuation date. This matters because the first-nearby contract can change during the averaging month.

Historical volatility is estimated from WTI futures returns available by the valuation date. The baseline uses the labelled continuous/front-month `CL=F` return series as the physical-measure proxy. A robustness check reconstructs a first-nearby return history contract by contract from CL settlements and excludes returns spanning contract switches.

3.2 | Date-ordered conditional pricing comparisons

We compare volatility inputs by repricing APOs using option-implied volatility recorded before each valuation date. Vendor first-publication times and the revision history of those records are not independently verified. Repricing uses the target-date futures curve, realized fixings, and discount factor; the volatility input alone comes from earlier option dates. Historical-volatility estimates follow the specified return-data cutoff. This is a date-ordered conditional pricing exercise, not a verified fully prospective forecast. The sample comprises 2,381 contract-date observations across seven expiries and 15 valuation dates.

The baseline integrates conditional prices over a volatility posterior estimated from an expanding return sample. Historical alternatives use rolling windows of 63, 126, or 252 returns, an exponentially weighted moving average (EWMA) with a 63-business-day half-life, and a horizon-matched Gaussian GARCH(1,1) forecast.

For GARCH, parameters are estimated from earlier returns; the valuation-date return updates conditional variance, which is then forecast over the remaining fixings. The comparison therefore uses some same-day information. We map these forecasts to the constant annualized volatility that matches the remaining average's variance under the one-factor pricing model. Full comparisons are in Online Supplement Section S1 and Table S1.

We obtain contract-level APO implied volatilities by inverting the Curran pricing approximation. For each expiry, the previous-day implied-volatility smile is estimated on the latest available earlier date. A second specification fits a smile using all earlier dates, giving more weight to recent observations. Both depend on moneyness and option type. Neither uses the target-date APO settlement to estimate volatility.

To assess the value of smile fitting, we also carry forward each contract's most recent implied volatility. We reprice with the valuation-date curve, discount factor, and fixing schedule. This same-contract benchmark covers 2,366 observations.

Pricing performance is summarized by mean error, mean absolute error (MAE), and root mean squared error (RMSE). Dependence across contracts observed on the same valuation date is preserved with valuation-date cluster bootstrap resampling.

3.3 | Cross-instrument validation

The within-APO comparison draws implied volatility from earlier observations of the same option family. We therefore add a cross-instrument check using standard monthly American WTI futures options (CME product LO) from Databento. We match LO option settlements to the underlying CL futures and invert prices using a Cox–Ross–Rubinstein futures-option tree.

For each APO target date, only LO observations from earlier dates are used. Implied-volatility smiles are fitted separately for the underlying CL futures contracts needed by the October-2026 APO fixing schedule. The resulting maturity-specific volatilities are mapped into the maintained scalar-volatility APO pricing model. We compare 253 APO contract-date observations over 10 target dates for which both external specifications can be evaluated within the moneyness range observed in earlier LO data. Online Supplement Section S4 and Table S4 report checks of the constant-volatility mapping and the number of steps in the Cox–Ross–Rubinstein tree.

This comparison asks whether the implied-volatility advantage also appears when volatility is estimated from another WTI option product; it does not identify a variance risk premium.

4 | RESULTS

4.1 | When does posterior integration matter?

Figure 1 shows when posterior integration affects point prices in the simulated settings. The largest differences occur with short estimation windows, high volatility, long maturities, few realized fixings, and pronounced price curvature. Across the 10 million synthetic histories, the maximum observed absolute difference is 0.1565 for a normalized $100 forward, and the largest within-cell 99th percentile is 0.1273. These extremes exceed the differences observed in the October-2026 sample. The sample maximum depends on the number of simulated histories; the within-cell percentile is less sensitive to rare simulated outcomes.

The second-order approximation tracks the simulated differences closely: the correlation between cell-level mean PI–PM differences and the curvature–variance term is 0.9995. Through-origin slopes are also close to one in the most extreme settings. These results support posterior variance and price curvature as the principal local determinants of the difference.

The PI–PM difference generally declines as the share of realized fixings increases. The relationship is not strictly monotone: in Panel C, the 252-day case initially rises because partial fixing changes both the remaining volatility exposure and local price curvature. The difference nevertheless tends toward zero as the averaging period ends.

4.2 | Posterior integration in the WTI sample

The October-2026 sample contains 310 contract-date observations across 12 valuation dates. Table 1 shows nearly identical pricing errors for PI and PM (RMSE 0.5177 and 0.5183, respectively). Their mean absolute price difference is only 0.000745 dollars per barrel.

The central 95% posterior interval for conditional model prices has mean width 0.2733 dollars per barrel, approximately 367 times the mean absolute PI–PM difference. This shows why a small integration correction does not imply a narrow posterior distribution of

When posterior integration matters

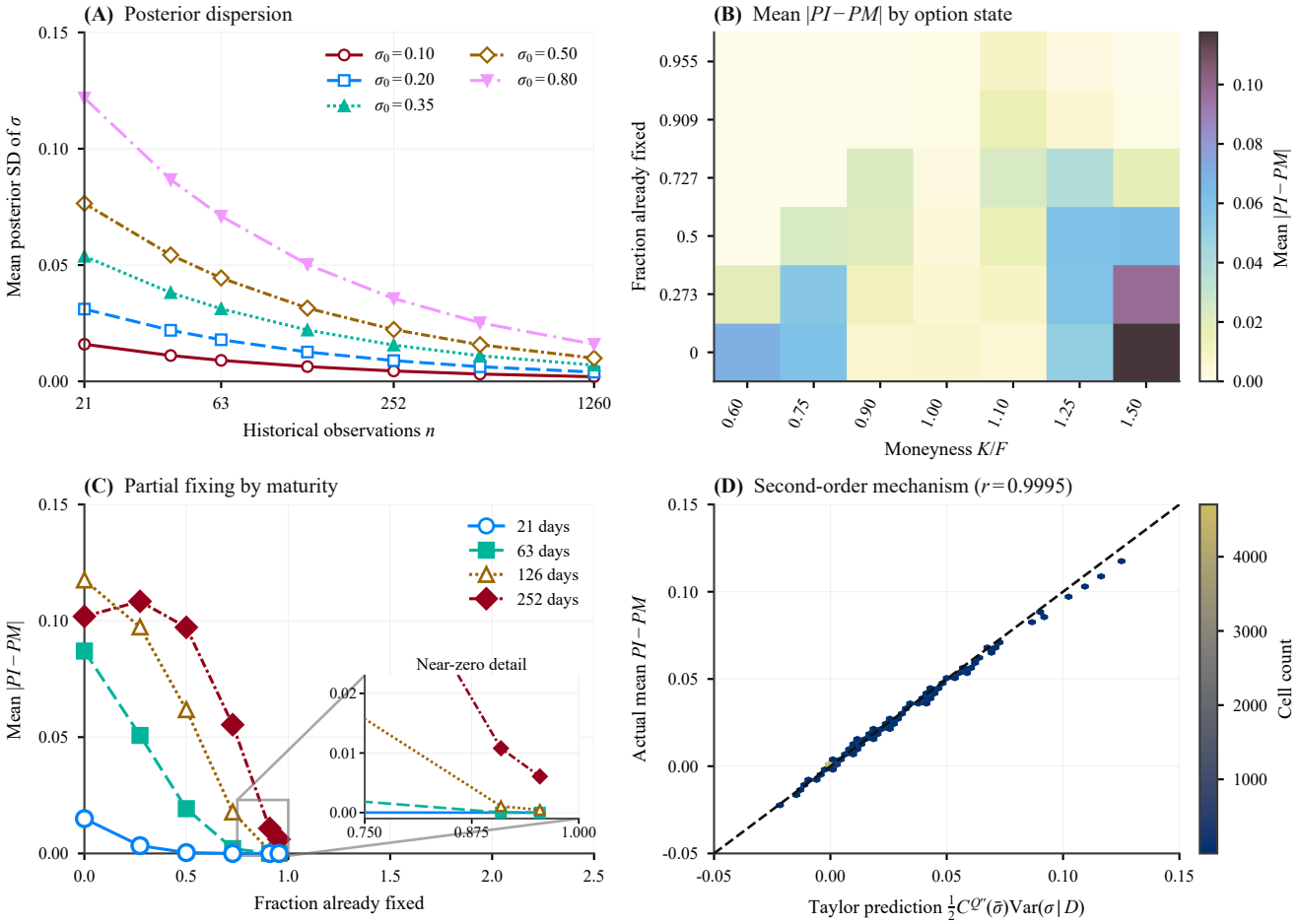


FIGURE 1 Posterior integration across 10 million synthetic return histories and WTI-like option states (root seed 2468). Panel A shows posterior dispersion by historical sample size. Panels B and C show the absolute PI–PM difference across moneyness and the realized-fixing fraction. Panel D compares mean PI–PM differences for 5,880 scenario cells with the second-order approximation in Equation (7); the cell-level correlation is 0.9995. The color bar shows the number of scenario cells per hexagonal bin.

model prices; it does not establish the economic value of that uncertainty. Both methods underprice October settlements on average.

4.3 | Historical versus option-implied volatility

Table 2 reports conditional pricing errors for 2,381 contract-date observations across 15 valuation dates. The historical PI benchmark has MAE 2.6187 and RMSE 3.4090, compared with 0.1075 and 0.1594 for the previous-day APO implied-volatility smile. The expanding earlier-date smile performs similarly. Both specifications reduce pricing errors for all seven expiries, with

TABLE 1 October-2026 WTI APO pricing and posterior price uncertainty. Pricing error is model price minus settlement price. All price quantities and errors are in U.S. dollars per barrel.

Quantity	N	Value
PI RMSE	310	0.5177
PM plug-in RMSE	310	0.5183
Mean $\|PI - PM\|$	310	0.000745
Mean 95% posterior price-interval width	310	0.2733

valuation-date cluster bootstrap intervals for the error differences below zero in each expiry (Figure 2).

Rolling-window, EWMA, and horizon-matched GARCH(1,1) benchmarks do not close the pricing-error gap (Online Supplement Section S1 and Table S1). In particular, GARCH gives MAE 2.6730 and RMSE 3.4441 on the same observations.

Panel B isolates 2,366 observations with an earlier implied volatility for the same contract. Carrying that volatility forward yields lower pooled errors than the fitted previous-day smile (MAE 0.0992 versus 0.1074). This benchmark shows that the advantage over historical volatility does not require smile fitting; it does not establish a general ranking of persistence and fitted-smile methods.

4.4 | Cross-instrument validation using vanilla-WTI options

The cross-instrument comparison replaces APO-implied volatility with estimates from standard monthly WTI futures options. Table 3 uses 253 October-2026 APO contract-date observations across 10 valuation dates. Historical-volatility posterior integration has MAE 0.4464 and RMSE 0.5562. A previous-day vanilla-WTI implied-volatility specification based on LO options lowers these errors to 0.1559 and 0.2599. The corresponding previous-day APO implied-volatility smile has MAE 0.1676 and RMSE 0.2686.

Valuation-date cluster bootstrap results support the improvement over historical volatility. Differences between LO-implied and APO-implied specifications are small and do not establish consistent superiority by MAE and RMSE. The cross-instrument test instead shows that the advantage persists when implied volatility comes from another WTI option family.

5 | ROBUSTNESS AND INTERPRETATION

Alternative physical-return assumptions do not materially change the comparison. A standardized Student-t model leaves the PI–PM difference small relative to historical-volatility pricing errors. Reconstructing first-nearby CL returns contract by contract gives a similar result: on the 2,381 date-ordered observations, historical PI has MAE 2.6468 and RMSE 3.4451, compared with 2.6187 and 3.4090 using the continuous/front-month return proxy. Further historical-volatility benchmarks appear in Online Supplement Section S1.

The small empirical PI–PM difference is not attributable to numerical noise. High-precision Monte Carlo, independently scrambled Sobol simulation, and Curran conditioning agree on the difference to a precision well below its observed magnitude. Simulation-based calibration provides a separate check on the posterior implementation. The supplement also reports a posterior-RMS plug-in sensitivity (Section S2), alternative mappings from LO-implied volatilities to one APO volatility, and Cox–Ross–Rubinstein tree-refinement checks (Section S4). These numerical checks do not establish an exact arbitrage-free representation of the volatility surface.

The market evidence has important limits. In the original 2,381-observation comparison, 2,319 option-date observations have zero recorded trading volume. The historical-versus-implied ranking remains under positive-volume and open-interest restrictions, but some expiry-level samples are small (Online Supplement Section S3). Moreover, the main pricing comparison contains only 15 valuation dates and the external comparison only 10. Publication timestamps and revision vintages of the original vendor records remain unverified. Valuation-date cluster resampling preserves dependence among contracts observed together, but cannot eliminate serial dependence across adjacent dates.

Finally, the one-factor constant-volatility model excludes richer futures-curve dynamics, stochastic volatility, jumps, and convenience yields. Its implied volatilities are model-equivalent pricing inputs, not uniquely identified risk-neutral diffusion parameters or direct estimates of a variance risk premium. The pricing-error differences apply to these inputs and this model; they do not identify physical and risk-neutral volatility components, trading gains, or the value of a hedging strategy.

6 | CONCLUSION

Posterior integration and the choice of volatility input affect WTI APO pricing in different ways. The synthetic experiment shows that PI and PM can diverge when the volatility posterior is dispersed and the pricing function has substantial curvature. In the October-2026 WTI sample, however, the PI–PM difference is small even though the posterior distribution of model prices remains much wider. Parameter-induced price dispersion should therefore not be inferred from the difference between two point estimates.

TABLE 2 Historical-volatility and prior option-implied volatility benchmarks. Panel A uses 2,381 date-ordered contract-date observations on 15 valuation dates. Panel B restricts the comparison to observations for which the same contract has a strictly prior implied volatility. Historical PI is the canonical Monte Carlo posterior-integrated price; scalar historical, GARCH, and implied-volatility inputs are repriced under the same Curran approximation. MAE and RMSE are in U.S. dollars per barrel.

Panel A: date-ordered conditional pricing comparison

Volatility input	*N*	Dates	MAE	RMSE
Historical PI	2,381	15	2.6187	3.4090
Rolling 63 returns	2,381	15	4.5201	5.6465
Horizon-matched GARCH(1,1)	2,381	15	2.6730	3.4441
Previous-day APO implied-volatility smile	2,381	15	0.1075	0.1594

Panel B: same-contract lagged-IV benchmark

Volatility input	*N*	Dates	MAE	RMSE
Historical PI	2,366	15	2.6279	3.4173
Same-contract lagged IV	2,366	15	0.0992	0.1403
Previous-day fitted APO smile	2,366	15	0.1074	0.1594

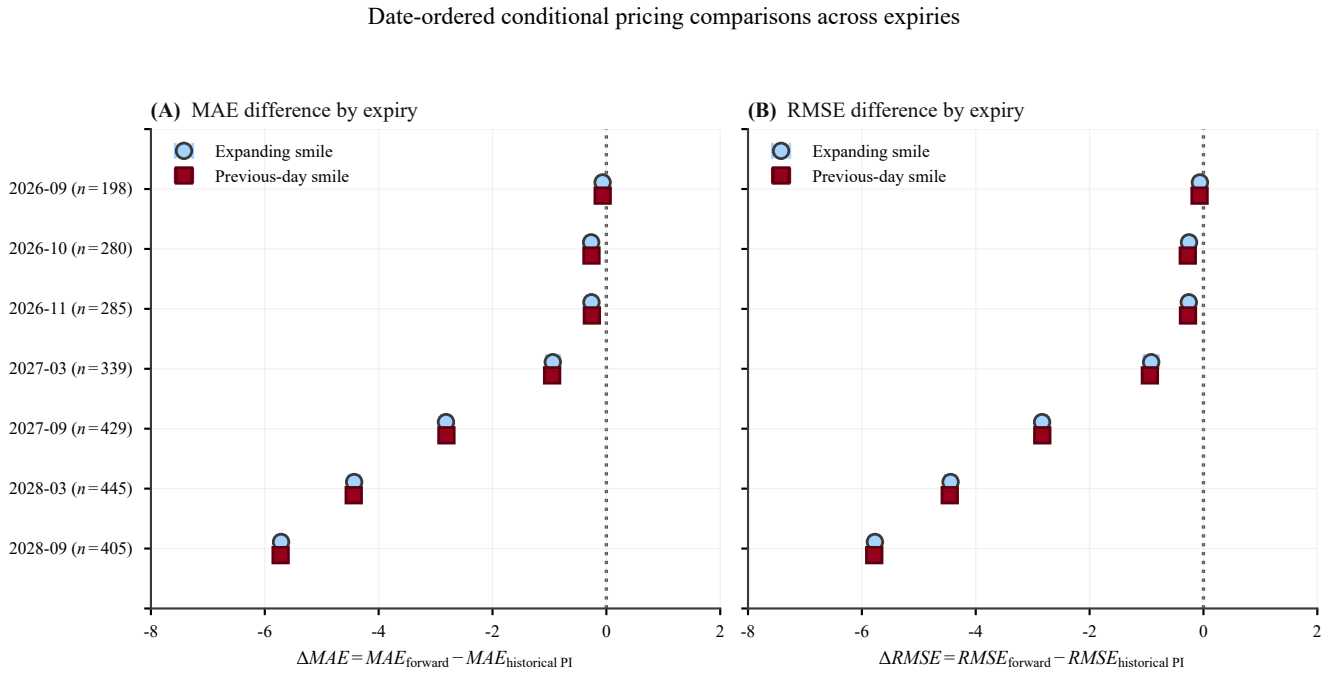


FIGURE 2 Date-ordered conditional pricing-error differences by APO expiry. Points compare the earlier-date APO implied-volatility specifications with the historical posterior-integrated benchmark; negative values indicate lower error for the implied-volatility specification. Horizontal bars are 95% percentile intervals from valuation-date cluster bootstrap resampling. All implied-volatility inputs are dated before the valuation date in the archived records; original publication times and subsequent revisions are not independently verified.

In the conditional APO pricing comparisons, earlier-date option-implied volatility gives lower errors against settlement-price proxies than the tested historical-volatility benchmarks. The result persists without cross-sectional smile fitting and when implied volatility is obtained from standard monthly WTI futures options. Within the maintained one-factor model and the limited number of valuation dates, the source of volatility information matters more for agreement

TABLE 3 Cross-instrument validation on 253 October-2026 APO contract-dates within observed prior LO moneyness support. Historical PI is the canonical Monte Carlo posterior-integrated price; the vanilla-WTI and APO implied-volatility inputs are repriced with the deterministic Curran approximation. All implied-volatility inputs use only earlier-date option information. MAE and RMSE are in U.S. dollars per barrel.

Volatility input	MAE	RMSE
Historical PI	0.4464	0.5562
Previous-day vanilla-WTI IV	0.1559	0.2599
Previous-day APO IV	0.1676	0.2686

with settlement-price proxies than posterior integration alone. These comparisons do not establish fully prospective performance or a general advantage over richer commodity models.

DATA AVAILABILITY

The WTI APO and CL histories used in the empirical application were obtained through a commercial Barchart subscription. End-of-day `Latest` prices are treated as settlement-price proxies, not independently verified transaction prices. The independent vanilla-WTI validation uses CME/NYMEX LO option and CL futures statistics accessed through Databento. Raw proprietary market-data files are not redistributed. The public repository provides code, query metadata, schemas, provenance records, diagnostics, hashes, and derived results consistent with the applicable data licenses.

CODE AVAILABILITY

The research code is organized as a Python package with tests and deterministic experiment entry points. The code and derived results are stored in a versioned repository. Its identifying URL is withheld from this double-anonymized review copy and can be supplied confidentially to the editor, subject to market-data licensing. The eventual submitted version will identify the exact computational snapshot.

PLAIN LANGUAGE SUMMARY

For WTI crude-oil average-price options, allowing for uncertainty about estimated volatility may change a model price only slightly. That does not imply the price is accurate or that volatility uncertainty is unimportant. We compare two ways to use the same volatility posterior and examine how much the choice of historical or earlier option-implied volatility affects conditional pricing errors.

AI-ASSISTED PREPARATION DISCLOSURE

OpenAI ChatGPT was used to assist with research software development, literature organization, and language editing. The author is responsible for the analysis, scientific claims, citations, and verification of the manuscript.

CONFLICT OF INTEREST

The author declares no conflict of interest.

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